LYRA HUANG

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EDUCATION

University of Virginia Expected May 2027
Charlottesville, Virginia | B.A. in Astronomy and B.A. in Computer Science
GPA: 3.754 / 4.00 (through Spring 2026)

RESEARCH EXPERIENCE

Undergraduate Researcher, Astronomy Senior Thesis 2026–Present
University of Virginia | Advisor: Prof. Phil Arras

- Investigating whether stellar magnetic fields can keep nanometer-scale charged dust grains near stars despite outward radiation pressure, and which conditions lead to escape, inward migration, or bounded orbits.
- Built Python/REBOUND orbit simulations combining gravity, radiation pressure, Poynting–Robertson drag, and electromagnetic forces; varied magnetic-field strength and polarity, grain size, and initial orbital radius.
- Revised radiation-pressure estimates for nanometer-scale grains using size-dependent efficiency Q_{pr} and β; compared trajectories to assess sensitivity to physical assumptions and model parameters.
- Checked numerical convergence, force magnitudes, and analytical cyclotron timescales; analyzed orbital trajectories, radial turning points, and escape behavior across parameter sweeps.

Undergraduate Research Tutorial | DREAMS Cosmological Simulations Spring 2026
University of Virginia | ASTR 4993 | Advisor: Prof. Paul Torrey

- Studied whether conditional generative models could reproduce distributions of galaxy properties from DREAMS simulations of galaxy formation under different physical conditions.
- Prepared DREAMS simulation datasets and adapted Python code for conditional normalizing flows, machine-learning models of probability distributions; ran training experiments on UVA's Rivanna cluster.
- Compared generated and simulated galaxy-property distributions across training configurations using visual and statistical diagnostics; documented the analysis in a March 2026 research report.

PROFESSIONAL EXPERIENCE

Software Intern, Software Department Summer 2026
VIA-tech | Shenzhen, China

- Annotation-data interoperability: Helped build conversions among YOLO, COCO, Pascal VOC, and Label Studio so computer-vision datasets could move between tools; used round-trip tests to check data consistency.
- Camera calibration: Contributed to AprilTag-assisted calibration and bird's-eye-view coordinate editing to relate camera images to spatial positions; supported camera-intrinsics import and MV3DT export.
- Multi-camera tracking: Contributed to Python tracking infrastructure using MQTT real-time data streams and timestamp synchronization to align observations across cameras.

SELECTED PROJECTS

Emberline | 2D Puzzle-Platformer Demo

Completed demo

Independent project | Godot 4 (GDScript) | Playable macOS and Windows builds

- Created a playable puzzle-platformer centered on carrying, throwing, and transferring a flame to activate light-sensitive and darkness-dependent mechanisms, bridges, and climbable vines.
- Implemented interactive level systems, player movement and climbing, checkpoints, hazards, audiovisual feedback, and a multi-stage level with a victory condition; included functional test scenes.

BeLie | 2D Puzzle Game Concept (Godot)

Fall 2026–Present

CS 4730, Computer Game Design | Team project, early development

- Co-developing a puzzle game concept in which players collect words through observation and combine them into belief statements that alter the world; one belief can be active at a time.
- Responsible for character design and map/terrain design during concept development, with Godot as the project engine.

MediCamp | Healthcare Visit Support App Prototype

HCI | Team of five | Project Website | Figma Prototype

- Managed milestones and team deliverables for a healthcare-support app prototype involving triage, appointment booking, navigation, and accessibility.
- Developed personas, task flows, wireframes, and a high-fidelity interactive Figma prototype; refined information architecture and interactions using critique and usability feedback.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, C, C++, SQL

Research and development: REBOUND, Rivanna HPC, Git/GitHub, Linux, Godot 4, GDScript, Figma, JUnit 5

Methods and tools: Numerical simulation, conditional normalizing flows, data visualization, camera calibration, computer vision annotation formats, MQTT, SQLite, Adobe Photoshop

TEACHING, OUTREACH & SERVICE

McCormick Observatory Public Nights Volunteer

Oct 2023–Present

University of Virginia

- Volunteer at public observing nights (approximately 0–2 events per semester, depending on scheduling); assist visitor flow and tours and explain astronomy concepts at events hosting around 120 attendees.

Astro Fun Club

Sep 2023–Present

University of Virginia

- Participate in astronomy observations, field activities, and science outreach.

HONORS & SELECTED COURSEWORK

Honors: Dean's List, University of Virginia (five semesters: Fall 2023, Spring 2024, Spring 2025, Fall 2025, Spring 2026).

Completed coursework: Human-Computer Interaction in Software Development; Astrophysics Research Methods; Observational Astronomy; Applied Linear Algebra; Data Structures and Algorithms; Introductory Cybersecurity; Discrete Mathematics.

Current coursework (Fall 2026): Artificial Intelligence; Machine Learning; Computer Game Design; Astronomy Senior Thesis; Minds, Machines, and Persons.